Temporal Dynamics of the Universal Somatic Field: Retarded Propagators, Transition Rates, and the Memory of Feeling

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Abstract

The Universal Somatic Field (USF) has been established as a scale-invariant field-theoretic architecture governing dynamics from quantum foam to the cosmic web. Previous papers have characterised the field's attractor structure, its Green's function identification, and its Lean 4 formal verification. The present paper completes the kinematic picture by developing the full time-dependent formulation. We derive the retarded Green's function $G_R(x, t; x', t')$ of the somatic field, characterise the relaxation time τ of attractor dynamics as a function of the spectral gap, and give a Wentzel–Kramers–Brillouin (WKB) estimate for the temporal barrier in emotional state transitions. We introduce the Somatic Memory Kernel $K(t - t')$ — the exponentially decaying influence of past field configurations on present dynamics — and show that it accounts for autobiographical memory, trauma re-experiencing, and the decay of conditioned responses within a single analytic framework. The Field-Modulated Hopfield Network (FM-HN) is extended to include an explicit time-dependent forcing current $J(x, t)$, enabling a treatment of therapeutic intervention as an optimal control problem on the field trajectory. Clinical implications include a field-theoretic derivation of why trauma processing is slower than trauma formation, and a quantitative account of the window of tolerance as a temporal bandwidth constraint on the retarded propagator. Core results are consistent with the Lean 4 formalisations in the companion lean-proofs-appendix paper.

Contents

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Introduction: Time in the Somatic Field

The Universal Somatic Field has been introduced and developed across a series of papers that have, with one exception, characterised it in the *stationary* regime: attractor basins, energy landscapes, phase transitions at critical thresholds. The exception is the quantum annealing experiment (Johnson, 2026c), which probed the *rate* at which the field crosses energy barriers — and found that the quantum pathway crosses them in fewer computational steps than any classical alternative.

But the stationary picture is incomplete. Real emotional life is not stationary. A person moves through emotional states; those states influence each other across time; past experiences leave traces that modulate present dynamics. The somatic field is not just a landscape — it is a trajectory through a landscape, and the trajectory has a velocity, an inertia, and a memory.

This paper develops the time-dependent formulation of the USF. The central object is the **retarded Green's function** $G_R(x, t;\, x', t')$: the response of the somatic field at position x and time t to a perturbation applied at position x' at the earlier time $t' < t$. The retarded Green's function is the causal propagator — it encodes only the past influence on the present, respecting the arrow of time. It is the natural object for describing memory, temporal integration, and the decay of emotional states.

The retarded propagator is not a new concept in physics; it is a standard tool in quantum field theory, classical electrodynamics, and the theory of open quantum systems. Its application to emotional dynamics is, however, new. The consequences are clinically significant: they give quantitative meaning to concepts such as the window of tolerance, the rate of trauma formation, the speed of therapeutic change, and the timescale of emotional memory consolidation.

1.1 Organisation of the Paper

Section 2 derives the retarded Green's function from the time-dependent somatic field equation. Section 3 introduces the Somatic Memory Kernel and shows how it unifies autobiographical memory, trauma re-experiencing, and conditioned response decay. Section 4 develops the WKB estimate of the temporal barrier in emotional state transitions. Section 5 extends the FM-HN to include time-dependent forcing and formulates therapeutic intervention as an optimal control problem. Section 6 draws clinical implications. Section 7 places the results in the context of the broader USF programme.

2

The Time-Dependent Somatic Field Equation

2.1 The Field Equation

The stationary somatic field equation (established in the companion papers) is:

$$(\nabla^2 + k^2)\,\Phi_{\mu\nu}(x) = -J_{\mu\nu}(x)$$

where $\Phi_{\mu\nu}$ is the somatic tensor field, k is the wavenumber (scale-dependent via the Zoom Operator Λ), and $J_{\mu\nu}$ is the source current encoding sensory input, motor output, and interoceptive signal.

The full time-dependent generalisation replaces the spatial Laplacian with the d'Alembertian:

$$\left(\frac{1}{v_s^2}\frac{\partial^2}{\partial t^2} - \nabla^2 + k^2\right)\Phi_{\mu\nu}(x,t) = -J_{\mu\nu}(x,t)$$

where v_s is the somatic field propagation velocity — the speed at which a perturbation in one part of the field influences another part. In neural tissue, v_s is determined by the conduction velocity of the electromagnetic field in the medium and the coupling constants of the somatic interaction; it is bounded above by the speed of light and below by the slowest neural conduction velocity.

The k^2 term acts as an effective mass: it prevents the field from propagating freely to arbitrarily large distances and introduces a characteristic length scale $\ell = 1/k$ beyond which field correlations decay exponentially. This length scale is the somatic correlation length — the spatial range over which different parts of the field influence each other. As the system approaches the consciousness threshold T_c, $k \to 0$ and $\ell \to \infty$: the correlation length diverges, signalling the onset of global integration.

2.2 The Retarded Green's Function

The retarded Green's function $G_R(x,t;\, x',t')$ satisfies:

$$\left(\frac{1}{v_s^2}\frac{\partial^2}{\partial t^2} - \nabla^2 + k^2\right) G_R(x,t;\, x',t') = \delta^{(3)}(x-x')\,\delta(t-t')$$

with the causal (retarded) boundary condition:

$$G_R(x,t;\, x',t') = 0 \quad \text{for } t < t'$$

The retarded condition is the mathematical statement that effects follow causes: the somatic field at (x,t) can be influenced only by sources at earlier times $t' < t$, never by future events.

In the homogeneous, isotropic case (uniform medium, no preferred direction), the retarded Green's function takes the form:

$$G_R(r,\tau) = \frac{v_s}{4\pi r}\, e^{-kv_s\tau}\, \delta(\tau - r/v_s)\, \theta(\tau)$$

where $r = |x - x'|$ is the spatial separation, $\tau = t - t' > 0$ is the elapsed time, and $\theta(\tau)$ is the Heaviside step function enforcing causality.

The key feature of this expression is the factor $e^{-kv_s\tau}$: the influence of a past perturbation decays exponentially in time, with decay constant kv_s. The decay rate is determined by the effective mass k — the same parameter that sets the spatial correlation length. A small k (near the consciousness threshold, or in a highly integrated system) means slow temporal decay: the field "remembers" perturbations for a long time. A large k (sub-threshold, highly localised) means rapid decay: the field forgets quickly.

2.3 The General Solution

Given the retarded Green's function, the general solution for the somatic field at time t is:

$$\Phi_{\mu\nu}(x,t) = \Phi^{(0)}_{\mu\nu}(x,t) + \int_{-\infty}^{t} dt' \int d^3x'\; G_R(x,t;\, x',t')\, J_{\mu\nu}(x',t')$$

where $\Phi^{(0)}_{\mu\nu}$ is the homogeneous solution (the field in the absence of external sources, evolving freely from initial conditions). The integral accumulates the influence of all past sources $J_{\mu\nu}(x',t')$ at earlier times $t' < t$, weighted by the retarded Green's function. This is the field's memory: it is the integral of past influences.

3

The Somatic Memory Kernel

3.1 Definition

The **Somatic Memory Kernel** $K(t-t')$ is the temporal part of the retarded Green's function, integrated over the spatial variables:

$$K(\tau) = \int d^3x\; G_R(x,\tau;\, 0,0) = K_0\, e^{-\tau/\tau_m}\, \theta(\tau)$$

where $\tau_m = 1/(kv_s)$ is the **somatic memory timescale** — the characteristic time over which the field retains the influence of past events.

The memory kernel is an exponentially decaying function of elapsed time. Recent events (small τ) have full influence; distant events have exponentially reduced influence. The timescale τ_m determines how far back in time the field "looks" when computing its present state.

3.2 Three Regimes of Memory

The memory timescale τ_m takes qualitatively different values in three phenomenologically distinct regimes:

Short-term somatic memory ($\tau_m \sim$ seconds to minutes): This is the timescale of conscious emotional experience. A felt emotion influences the field for this duration before decaying if no further reinforcement occurs. The corresponding wavenumber k is in the range associated with the cortical-limbic coupling frequency band (theta/alpha range, 4–12 Hz).

Medium-term somatic memory ($\tau_m \sim$ hours to days): This is the timescale of mood and emotional state. A disrupted sleep, a difficult conversation, a sustained stressor — these perturb the field on this timescale. The corresponding k is smaller, the spatial correlation length longer; these states involve wider-spread field configurations.

Long-term somatic memory ($\tau_m \sim$ months to years, or indefinitely): This is the timescale of autobiographical memory and, in the pathological case, of trauma well structures. Long-term somatic memory corresponds to near-zero k: the field is near the critical point, the correlation length is very large, and past perturbations decay extraordinarily slowly — in the limit $k \to 0$, the memory is permanent.

3.3 Trauma Re-experiencing as Memory Kernel Resonance

The characteristic symptom of PTSD and complex PTSD — the involuntary re-experiencing of traumatic events as if they were present — receives a precise account in the memory kernel framework. A traumatic event creates a very deep, narrow well in the energy landscape AND a very long memory timescale: the k parameter of the trauma-well basin is anomalously small, meaning the kernel decay constant $\tau_m = 1/(kv_s)$ is anomalously large.

Re-experiencing occurs when a present stimulus (a sensory trigger) generates a source current $J(x,t)$ that overlaps with the memory kernel of the trauma: the integral $\int K(t-t')J(t')\,dt'$ yields a large response because the kernel has not decayed to zero at the relevant times. The field "replays" the past event because the memory kernel ensures the past event still has appreciable weight in the field's present configuration.

This gives a quantitative criterion for trauma severity: the depth of the trauma well determines the barrier height (relevant for escape rate, as in the existing WKB analysis); the k-value of the trauma-well basin determines the memory timescale (relevant for re-experiencing frequency and vividness).

3.4 Conditioned Response Decay

The extinction of conditioned responses — the gradual weakening of an emotional response to a stimulus after repeated unreinforced presentations — is the exponential decay of the memory kernel in the absence of reinforcement. The field's response to the conditioned stimulus decays as $e^{-\tau/\tau_m}$; after a time of order several τ_m, the response has substantially attenuated.

This gives a field-theoretic account of exposure therapy: repeated presentation of the conditioned stimulus without the unconditioned stimulus (the trauma) allows the memory kernel contribution to decay. The therapy does not erase the memory — it reduces K_0, the initial amplitude of the kernel, through the accumulation of unreinforced presentations that progressively lower the well depth.

4

WKB Estimate of the Temporal Barrier

4.1 Transition Rate in the Time Domain

The stationary WKB analysis gives the barrier *height* in the energy landscape. The time-dependent formulation adds the temporal dimension: how long does a transition between attractor basins take?

In the overdamped Kramers escape problem (which applies when the field's effective temperature is low and the dynamics are diffusive rather than ballistic), the mean first-passage time $\langle T \rangle$ from attractor basin A to attractor basin B is:

$$\langle T \rangle = \frac{2\pi}{\omega_A \omega_B} e^{\Delta V/D}$$

where ω_A is the frequency of oscillation at the bottom of basin A (the curvature of the well), ω_B is the magnitude of the imaginary frequency at the saddle point between A and B (the curvature at the transition state), ΔV is the barrier height (the energy difference between the saddle and the bottom of A), and D is the diffusion coefficient (proportional to the field temperature T_{field}).

In the somatic field, this formula gives the expected time to make a spontaneous emotional transition from state A to state B without external perturbation.

4.2 Asymmetry of Formation and Dissolution

The Kramers formula immediately explains a clinically well-established asymmetry: **trauma formation is much faster than trauma dissolution**. The reason is the asymmetry of the barrier structure:

- **Trauma formation**: The traumatic event delivers a large, brief forcing current $J_{\text{trauma}}(x,t)$ that drives the field over the barrier from the healthy attractor into the trauma well *from above* — the full barrier height is not required because the external forcing provides most of the needed energy. The formation time is of order the duration of the traumatic event: seconds to minutes.
- **Trauma dissolution**: Escape from the trauma well requires crossing the full barrier from *below* (from inside the well), without the benefit of a large external forcing. The mean first-passage time grows exponentially with the barrier height. For a deep trauma well ($\Delta V \gg D$), this time can be astronomically large without therapeutic intervention.

This is not a failure of the person trapped in the trauma well. It is the physics: a ball dropped into a deep pit takes far less energy to drop than to climb back out. The somatic field is following its equations.

4.3 The Window of Tolerance as Temporal Bandwidth

The clinical concept of the "window of tolerance" — the range of field temperatures within which therapeutic processing of traumatic material is possible — receives a temporal reformulation in the retarded propagator framework.

The window of tolerance is the range of temperatures $[T_{\text{min}}, T_{\text{max}}]$ such that:

1. $T > T_{\text{min}}$: The relaxation time $\tau_{\text{relax}} = 1/(\omega_A^2/\gamma)$ (where γ is the damping coefficient) is short enough that the field can visit the trauma-adjacent region and return to safety within the therapeutic session. Below T_{min} (frozen state), the relaxation time is too long: the field enters the vicinity of the trauma well but cannot return to the regulated basin within the session window.

2. $T < T_{\text{max}}$: The mean first-passage time $\langle T \rangle$ from the current regulated attractor to the trauma well is long enough that the field does not spontaneously cascade into the trauma well during the session. Above T_{max} (flooding state), the field is too hot: it crosses the barrier into re-traumatisation without therapeutic benefit.

The window of tolerance is therefore a temporal bandwidth: a range of field temperatures within which the retarded propagator ensures that the field can approach and retreat from the trauma-adjacent region on therapeutic timescales. This gives a quantitative grounding for the standard clinical instruction to "titrate" trauma processing — to regulate the field temperature so that processing occurs within the window.

5

Therapeutic Intervention as Optimal Control

5.1 The Forced Field Equation

The full time-dependent field equation with external therapeutic forcing is:

$$\left(\frac{1}{v_s^2}\frac{\partial^2}{\partial t^2} - \nabla^2 + k^2\right)\Phi(x,t) = -J_{\text{endo}}(x,t) - J_{\text{therapy}}(x,t)$$

where J_{endo} is the endogenous source current (the body's own sensory and interoceptive signals) and J_{therapy} is the external therapeutic forcing current — the physical effect of the therapist's interventions on the somatic field.

The therapeutic current $J_{\text{therapy}}(x,t)$ is non-zero during therapy sessions and zero between sessions. Different therapeutic modalities correspond to different spatial and temporal profiles of J_{therapy}:

- **Somatic therapies** (SE, SP, EMDR): J_{therapy} has large amplitude at somatic field frequencies (low frequency, body-centred), high spatial correlation within the somatic register, and rapid temporal variation matched to the client's biological rhythms.
- **Cognitive therapies** (CBT, schema): J_{therapy} has large amplitude at cortical frequencies (higher frequency, language-centred), centred on the cortical subspace, and slower temporal variation.
- **Pharmacological interventions**: J_{therapy} modifies the field parameters themselves — the wavenumber k, the damping coefficient γ, or the field temperature T_{field} — rather than applying a forcing current. SSRIs, for example, raise the field temperature by increasing serotonergic coupling, making the field more willing to explore the landscape.

5.2 The Optimal Control Problem

Given a current field configuration $\Phi(x,t_0)$ (assessed at the start of therapy) and a target configuration $\Phi^\star(x)$ (the healthy regulated attractor), the optimal control problem is: find the therapeutic current $J_{\text{therapy}}(x,t)$ that drives the field from $\Phi(x,t_0)$ to $\Phi^\star(x)$ in minimum time, subject to the constraint that the field remains within the window of tolerance throughout.

This is a standard variational problem with state constraints, and it can be solved by the Pontryagin minimum principle. The solution gives the optimal therapeutic trajectory: the sequence of interventions, each with its appropriate amplitude and timing, that achieves the therapeutic goal most efficiently.

The formal solution is beyond the scope of this paper (it requires specifying the energy landscape, the window-of-tolerance constraints, and the admissible set of therapeutic currents). However, the framework establishes that such an optimal solution *exists*, is *computable in principle*, and provides a *principled criterion* for evaluating any proposed therapeutic approach: does it approximate the optimal control trajectory, or does it systematically deviate from it?

5.3 Why Somatic Entry Is Faster: A Formal Account

The claim made in the clinical companion papers (Johnson, 2026a) — that somatic entry to traumatic material is more efficient than cognitive entry — now has a formal account. The efficiency difference arises from the spatial structure of J_{therapy}:

Somatic interventions apply J_{therapy} directly to the somatic-limbic subspace of the field, which is the subspace containing the trauma well. The forcing current reaches the trauma-well basin directly, without passing through the EC (Emotional Core) junction.

Cognitive interventions apply J_{therapy} to the cortical subspace. Reaching the somatic-limbic basin from the cortical subspace requires traversing the EC junction, which is the point of maximum decoupling in CPTSD presentations. The effective coupling between the cortical forcing and the trauma well is proportional to the EC coupling constant κ_{EC} — which is anomalously small in CPTSD by definition.

The efficiency ratio is therefore approximately κ_{EC}^{-1}: somatic therapy is κ_{EC}^{-1} times more efficient at perturbing the trauma well per unit of therapeutic effort, compared to cognitive therapy. For severely decoupled CPTSD presentations (small κ_{EC}), this ratio can be large — consistent with the clinical observation that complex trauma often requires body-based approaches to access what decades of talking cannot reach.

6

The Temporal Somatic Field Across Scales

6.1 Developmental Timescales

The memory kernel framework applies across the developmental lifespan. The somatic field of a developing organism has a time-dependent wavenumber $k(t)$ that decreases across development: the infant's field begins with large k (short memory, high k, rapid emotional state transitions — consistent with the rapid emotional cycling of early infancy) and develops toward smaller k (longer memory, greater emotional stability, broader somatic integration).

The critical developmental events — secure attachment formation, the emergence of language and narrative memory, the consolidation of identity — each correspond to specific reductions in k: transitions to longer memory timescales at which new categories of experience become possible.

The pre-verbal manifold (Johnson, 2026e) — the field structure present before language acquisition — is characterised by large k (rapid decay, short memory), which explains why pre-verbal memories are not accessible to verbal recall: the memory kernel of the pre-verbal field has decayed to zero by the time the verbal apparatus is present to encode it. The somatic field retained the event, but in a form that does not project onto the language channel.

6.2 Geological and Astrophysical Timescales

The same retarded propagator applies at geological and astrophysical scales, with the scale-appropriate velocity v_s and wavenumber k. At geological scale ($\sigma = 10$ in the Zoom Operator notation), the somatic field describes seismic wave propagation and tectonic dynamics. The memory kernel at this scale has timescale $\tau_m \sim 10^3$ to 10^6 years — the timescale over which geological stress distributions encode the history of past tectonic events. This is the field-theoretic account of geological memory: rock strata remember (Johnson, 2026d).

At cosmological scale ($\sigma = 19$–20), the retarded propagator is the cosmological Green's function — the propagator for perturbations in the early universe, whose memory kernel timescale is the Hubble time $\sim 10^{10}$ years. The cosmic microwave background is the long-memory trace of quantum fluctuations in the very early universe: an exponentially decayed but still detectable somatic memory of the universe's infancy.

7

Implications and Open Questions

7.1 Measurable Predictions

The temporal dynamics framework makes several predictions that are, in principle, directly testable with existing technology.

Memory timescale from EEG spectral width: The somatic memory timescale $\tau_m = 1/(kv_s)$ is inversely proportional to the effective wavenumber k. In neural tissue, k is related to the centre frequency and bandwidth of the dominant EEG oscillation: broader spectral peaks correspond to larger effective k (shorter memory), narrower peaks to smaller k (longer memory). A systematic study of EEG spectral width in populations with different trauma histories would test the prediction that longer trauma history correlates with narrower EEG spectral peaks (smaller k, longer memory timescale).

Transition rate asymmetry: The Kramers formula predicts that the ratio of trauma-formation time to trauma-dissolution time grows exponentially with barrier height. An empirical study of how long clinically significant traumatic events take to produce lasting field changes, compared with how long therapeutic dissolution of equivalent changes requires, would test the predicted asymmetry and constrain the barrier heights involved.

Window of tolerance as frequency window: The temporal bandwidth of the window of tolerance should correspond to a specific frequency range in the physiological data. Heart rate variability, skin conductance, and respiratory rate all have frequency content that reflects the field temperature. The prediction is that therapeutic sessions are most productive when physiological frequency indicators are within a specific band — and that this band is the same band predicted by the Kramers formula for the given barrier height.

7.2 The Time Variable in the Lean 4 Formalisation

The spatial (stationary) aspects of the USF have been formally verified in Lean 4 (Johnson, 2026b). The time-dependent formulation introduces new proof obligations. The retarded boundary condition (causality) should be stated as a Lean 4 theorem; the exponential decay of the memory kernel should follow as a corollary; the Kramers mean first-passage time formula should be derived from the field equation in the overdamped limit.

The formal statement of causality — that $G_R = 0$ for $t < t'$ — is precisely a **dependent type constraint** in the Lean 4 type system. In the type-theoretic reading, the retarded propagator has the type:

$$G_R : (t\, t' : \text{Time}) \to (t' < t) \to \text{Space} \to \text{Space} \to \text{Field}$$

The inequality $t' < t$ is a proof argument — a term of type `Prop` that must be supplied at every call site. The Lean 4 kernel enforces causality at compile time: any use of the propagator that does not supply a proof of $t' < t$ is a type error. The temporal arrow of time is not a convention but a structural constraint woven into the type signature of the propagator.

This is the type-theoretic completion of the USF's kinematic picture. The spatial Σ-type (the soma-field as a dependent sum over scale levels, established in `ScaleUniverse.lean`) is joined by the temporal dependent type (the retarded propagator as a function that takes a causality proof). Together they give the full USF type:

$$\text{USF} \;\equiv\; \sum_{\sigma:\text{Scale}_{20}} (\text{Substrate}(\sigma) \;\times\; G_R(\sigma))$$

where $G_R(\sigma)$ is the retarded propagator at scale σ, carrying the causality constraint as a proof argument.

7.3 The Unified Kinematic Picture

The temporal dynamics paper completes the kinematic picture of the Universal Somatic Field. The full specification of the field now includes:

Property	Object	Key formula
Spatial structure	Attractor landscape	$V[\Phi] = -\frac{1}{2}\Phi \cdot W \cdot \Phi$ (Hopfield)
Spatial propagation	Green's function $G(x, x')$	$(\nabla^2 + k^2)G = \delta^3(x - x')$
Temporal propagation	Retarded propagator $G_R(x, t; x', t')$	$(\Box + k^2)G_R = \delta^4(x - x')$, $G_R = 0$ for $t < t'$
Memory	Memory kernel $K(\tau)$	$K_0 e^{-\tau/\tau_m}$
Transition rate	Kramers formula	$\langle T \rangle = \frac{2\pi}{\omega_A \omega_B} e^{\Delta V/D}$
Phase transition	Consciousness threshold	Spectral gap opens at T_c
Scale invariance	Zoom Operator Λ	$k \mapsto k(\sigma)$ for $\sigma \in \{0, \dots, 20\}$

The field is now fully specified in both space and time. The attractor landscape tells us where the field goes; the temporal dynamics tell us how fast it gets there, how long it stays, and what traces it leaves behind.

8

Conclusion

The temporal dynamics of the Universal Somatic Field are governed by the retarded Green's function, the Somatic Memory Kernel, and the Kramers transition rate formula. Together, these three objects provide a complete kinematic description of emotional dynamics: where the field can go (the attractor landscape), how long it takes to get there (the Kramers rate), how long it remembers where it has been (the memory kernel), and how an external therapist can guide it most efficiently (the optimal control formulation).

The framework resolves several clinical puzzles: - Why trauma forms faster than it dissolves (asymmetric barrier crossing) - Why somatic therapies are more efficient than cognitive ones for complex PTSD (somatic forcing bypasses the decoupled EC junction) - Why pre-verbal memories are inaccessible to verbal recall (the pre-verbal memory kernel has decayed to zero by the time the verbal apparatus is present) - What the window of tolerance is, formally (a temporal bandwidth constraint on the retarded propagator)

And it opens new empirical programmes: - EEG spectral width as a proxy for somatic memory timescale - Transition rate asymmetry as a quantitative test of the Kramers formula - Therapeutic session physiology as a test of the window-of-tolerance bandwidth prediction

The somatic field has a past. The retarded propagator carries it. The present is the integral of everything that has happened, weighted by how long ago it happened and how much it mattered.

9

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